Contents

1. The Natural Numbers

Exercise 1.1

Prove using mathematical induction that for all positive integers n,

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

Proof. Let X be the set of all natural numbers for which the formula above holds true. $1 \in X$ because

$$1 = \frac{1(1+1)}{2}.$$

Now assume that $n > 1$, and that $k \in X$ for all $k < n$. We know from our induction hypothesis that the formula holds for $n - 1$, so

$$1 + 2 + 3 + \cdots + (n-1) = \frac{(n-1)n}{2}.$$

We add n to both sides to obtain

$$\begin{aligned} 1 + 2 + 3 + \cdots + (n-1) + n &= \frac{(n-1)n}{2} + n \\ &= \frac{n^2 - n + 2n}{2} \\ &= \frac{n^2 + n}{2} \\ &= \frac{n(n+1)}{2}. \end{aligned}$$

Therefore, $n \in X$. It follows from the principle of mathematical induction that $X = \mathbb{N}$. □

Exercise 1.2

Prove using mathematical induction that for all positive integers n,

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(2n+1)(n+1)}{6}.$$

Proof. Let X be the set of natural numbers from which the formula above holds. $1 \in X$ because

$$1^2 = \frac{1(2 \cdot 1 + 1)(1+1)}{6}.$$

Now suppose that $n > 1$, and that $k \in X$ for all $k < n$. In particular, $n - 1 \in X$ so

$$1^2 + 2^2 + \cdots + (n-1)^2 = \frac{n(n-1)(2n-1)}{6}.$$

Adding n^2 to both sides, we deduce that

$$\begin{aligned}1^2 + 2^2 + \cdots + (n-1)^2 + n^2 &= \frac{n(n-1)(2n-1)}{6} + n^2 \\ &= \frac{2n^3 - 3n^2 + n + 6n^2}{6} \\ &= \frac{2n^3 + 3n^2 + n}{6} \\ &= \frac{n(2n^2 + 3n + 1)}{6} \\ &= \frac{n(2n+1)(n+1)}{6}.\end{aligned}$$

So $n \in X$, which closes the induction. □

Exercise 1.3

Assuming the triangle inequality ($|x+y| \leq |x| + |y|$) is true, show that $|x_1 + x_2 + \cdots + x_n| \leq |x_1| + |x_2| + \cdots + |x_n|$ for all $n \in \mathbb{N}$.

Proof. Let X be the set of all natural numbers n such that $|x_1+x_2+\cdots+x_n| \leq |x_1|+|x_2|+\cdots+|x_n|$. $1 \in X$ and it's given that $2 \in X$. Suppose that $k \in X$ for all $k < n$ where $n > 2$. In particular, that means $n-1 \in X$. Therefore, $|x_1 + x_2 + \cdots + x_{n-1}| \leq |x_1| + |x_2| + \cdots + |x_{n-1}|$. We add $|x_n|$ to both sides to obtain

$$|x_1 + x_2 + \cdots + x_{n-1}| + |x_n| \leq |x_1| + |x_2| + \cdots + |x_{n-1}| + |x_n|.$$

Let $y = x_1 + x_2 + \cdots + x_{n-1}$, then the left-hand side of the inequality above becomes $|y| + |x_n|$. Because $2 \in X$, we know $|y + x_n| \leq |y| + |x_n|$. Therefore, $|y + x_n| \leq |x_1| + |x_2| + \cdots + |x_n|$, which closes the induction and shows that $X = \mathbb{N}$. □

Exercise 1.4

Given a positive integer n, recall that $n! = 1 \cdot 2 \cdot 3 \cdots n$ (this is read as n **factorial**). Provide and inductive definition for $n!$. (It is customary to actually use this definition at $n = 0$, setting $0! = 1$.)

Proof.

$$n! = n(n-1)!$$

□

Exercise 1.5

Prove that $2^n < n!$ for all $n \geq 4$.

Proof. Let X be the set of all natural numbers n such that $2^n < n!$. Define $S := X \cup \{1, 2, 3\}$, we show that $S = \mathbb{N}$ by induction. By definition, $1 \in S$. Now suppose that $k \in S$ for all $k < n$ where $n > 1$. If $n = 2$ or $n = 3$, then by definition $n \in S$. If $n = 4$, then $2^4 = 16$ and $4! = 24$. Clearly $16 < 24$, therefore $n \in X$. Now we can solely focus on the cases where $n > 4$. We know $n - 1 \in S$, so $2^{n-1} < (n-1)!$ or $n - 1 \in \{1, 2, 3\}$. But $n > 4$ so $n - 1 \notin \{1, 2, 3\}$. So it must be the case that $2^{n-1} < (n-1)!$. Hence, $2^n < 2(n-1)!$. But $2(n-1)! < n(n-1)! = n!$. Therefore, $2^n < n!$. This closes the induction and shows that $S = \mathbb{N}$. $\square$

Exercise 1.6

Prove that for all positive integers n,

$$1^3 + 2^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2}\right)^2.$$

Proof. Let X be the set of all natural numbers n such that the equality above holds. We show that $X = \mathbb{N}$ by induction. Because

$$1^3 = \left(\frac{1(1+1)}{2}\right)^2,$$

we know that $1 \in X$. Now suppose that $k \in X$ for all $k < n$ where $n > 1$. In particular, $n - 1 \in X$. Therefore,

$$1^3 + 2^3 + \cdots + (n-1)^3 = \left(\frac{n(n-1)}{2}\right)^2.$$

Adding n^3 to both sides, we have

$$\begin{aligned}
1^3 + 2^3 + \cdots + (n-1)^3 + n^3 &= \left(\frac{n(n-1)}{2}\right)^2 + n^3 \\
&= \frac{n^2(n-1)^2 + 4n^3}{4} \\
&= \frac{n^2(n^2-2n+1) + 4n^3}{4} \\
&= \frac{n^4 - 2n^3 + n^2 + 4n^3}{4} \\
&= \frac{n^4 + 2n^3 + n^2}{4} \\
&= \frac{n^2(n^2+2n+1)}{2^2} \\
&= \frac{n^2(n+1)^2}{2^2} \\
&= \left(\frac{n(n+1)}{2}\right)^2.
\end{aligned}$$

This closes the induction and proves that $X = \mathbb{N}$. □

Exercise 1.7

Prove the familiar **geometric progression** formula. Namely, suppose that a and r are real numbers with $r \neq 1$. Then show that

$$a + ar + ar^2 + \cdots + ar^{n-1} = \frac{a - ar^n}{1-r}.$$

Proof. Let $S = a + ar + ar^2 + \cdots + ar^{n-1}$.

$$\begin{aligned}
S &= a + r(a + ar^2 + \cdots + ar^{n-2}) \\
&= a + r(S - ar^{n-1}) \\
&= a + rS - ar^n
\end{aligned}$$

Thus, $(1-r)S = a - ar^n$ and

$$S = \frac{a - ar^n}{1-r}.$$

□

Exercise 1.8

Prove that for all positive integers n,

$$\frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\cdots+\frac{1}{n(n+1)}=\frac{n}{n+1}.$$

Proof. Let X be the set of natural numbers n such that the equality above holds. One can quickly verify that $1 \in X$. Now suppose that $k \in X$ for all $k < n$ where $n > 1$. In particular, $n-1 \in X$. Hence,

$$\frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\cdots+\frac{1}{n(n-1)}=\frac{n-1}{n}.$$

Adding $\frac{1}{n(n+1)}$ to both sides and performing some simple algebra, we yield

$$\begin{aligned}\frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\cdots+\frac{1}{n(n+1)} &= \frac{n-1}{n}+\frac{1}{n(n+1)}\\ &= \frac{(n-1)(n+1)+1}{n(n+1)}\\ &= \frac{n^2-1+1}{n(n+1)}\\ &= \frac{n^2}{n(n+1)} && = \frac{n}{n+1}.\end{aligned}$$

This closes the induction and shows that $X = \mathbb{N}$. $\square$

Exercise 1.9

By trial and error, try to find the smallest positive integer expressible as $12x + 28y$, where x and y are allowed to be any integers.

Proof. $x = -2$ and $y = 1$ gives 4. $\square$

Exercise 1.10

Use mathematical induction to show that the complete graph of n points has exactly $n(n-1)/2$ lines.

Proof. Let X be the set of natural numbers n for which the formula above works. We show that $S := X \cup \{1\}$ is equal to $\mathbb{N}$ by induction. By definition, $1 \in S$. Suppose that $k \in S$ for all $k < n$ where $n > 1$. Assume $n = 2$, then $2(2-1)/2 = 1$, so $2 \in S$. Assume $n > 2$, we know from our induction hypothesis that $n-1 \in S$, but $n-1 > 1$ and thus cannot be an element of $\{1\}$. So $n-1 \in X$. So the complete graph of $n-1$ has exactly $(n-1)(n-2)/2$ lines. Now add another point to the graph. That point will form a line with every other point (of which there are $n-1$). So

$n-1$ new lines will be formed. Hence, the complete graph of n will have $(n-1)(n-2)/2+(n-1)$ lines. Using some algebra,

$$\begin{aligned}\frac{(n-1)(n-2)+2(n-1)}{2} &= \frac{(n-1)(n-2+2)}{2} \\ &= \frac{n(n-1)}{2}.\end{aligned}$$

This closes the induction and shows that $S=\mathbb{N}$, as desired. □

Exercise 1.11

Consider the sequence $\{a_n\}$ defined inductively as follows:

$$a_1 = a_2 = 1, \quad a_{n+2} = 2a_{n+1} - a_n.$$

Use mathematical induction to prove that $a_n = 1$, for all natural numbers n.

Proof. Let X be the set of all natural numbers n such that $a_n = 1$. By the sequence's definition, $1 \in X$ and $2 \in X$. Now suppose that $k \in X$ for all $k < n$ for some $n > 2$. So $a_{n-1} = 1$ and $a_{n-2} = 1$. By definition, $a_n = 2a_{n-1} - a_{n-2} = 2(1) - 1 = 1$. Thereby, $n \in X$ and $X = \mathbb{N}$. □

Exercise 1.12

Consider the sequence $\{a_n\}$ defined inductively as follows:

$$a_1 = 5, \quad a_2 = 7, \quad a_{n+2} = 3a_{n+1} - 2a_n.$$

Use mathematical induction to prove that $a_n = 3 + 2^n$, for all natural numbers n.

Proof. Let X be the set of all $n \in \mathbb{N}$ such that $a_n = 3 + 2^n$. Trivially, $1 \in X$ and $2 \in X$. Now suppose that $k \in X$ for all $k < n$ where $n > 2$. Thus, $a_{n-1} = 3 + 2^{n-1}$ and $a_{n-2} = 3 + 2^{n-2}$. By definition, it follows that $a_n = 3a_{n-1} - 2a_{n-2}$. So $a_n = 3(3 + 2^{n-1}) - 2(3 + 2^{n-2})$.

$$\begin{aligned}a_n &= 3(3 + 2^{n-1}) - 2(3 + 2^{n-2}) \\ &= 9 + 3 \cdot 2^{n-1} - 6 - 2^{n-1} \\ &= 3 + 2^n.\end{aligned}$$

This shows that $n \in X$, and thus $X = \mathbb{N}$. □

Exercise 1.13

Consider the Fibonacci sequence $\{a_n\}$.

(a) Use mathematical induction to prove that

$$a_{n+1}a_{n-1} = (a_n)^2 + (-1)^n.$$

(b) Use mathematical induction to prove that

$$a_n = \frac{(1+\sqrt{5})^n - (1-\sqrt{5})^n}{2^n\sqrt{5}}.$$

Proof. (a) Let X be the set of all $n \in \mathbb{N}$ such that $a_{n+1}a_{n-1} = (a_n)^2 + (-1)^n$. We show that $S := X \cup \{1\} = \mathbb{N}$ by induction. By definition, $1 \in S$. Now suppose that $k \in S$ for all $k < n$ where $n > 1$. If $k = 2$, then $a_3a_1 = (2)(1) = 2$ and $(a_2)^2 + (-1)^2 = 1^2 + 1 = 2$. Therefore, $2 \in S$. Now suppose that $n > 2$, then $n-1 \in S$ but $n-1 \neq 1$, therefore $n-1 \in X$. By our induction hypothesis, $a_na_{n-2} = (a_{n-1})^2 + (-1)^{n-1}$.

$$\begin{aligned}
a_{n+1}a_{n-1} &= (a_{n-1} + a_n)(a_n - a_{n-2}) \\
&= a_{n-1}a_n - a_{n-1}a_{n-2} + (a_n)^2 - a_na_{n-2} \\
&= a_{n-1}(a_n - a_{n-2}) + (a_n)^2 - ((a_{n-1})^2 + (-1)^{n-1}) \\
&= (a_{n-1})^2 + (a_n)^2 - (a_{n-1})^2 - (-1)^{n-1} \\
&= (a_n)^2 - (-1)^{n-1} \\
&= (a_n)^2 + (-1)^n.
\end{aligned}$$

This shows that $n \in X$, which implies $n \in S$, which implies $S = \mathbb{N}$.

(b) First of all, we define

$$\phi := \frac{1+\sqrt{5}}{2} \qquad \text{the golden ratio.}$$

One can confirm using simple algebra that

$$\begin{aligned}
2 - \phi &= (1-\phi)^2 \\
\phi + 1 &= \phi^2 \\
1 - \phi &= \frac{1-\sqrt{5}}{2}
\end{aligned}$$

Using the definition and properties of ϕ, we can rewrite the formula as

$$a_n = \frac{\phi^n - (1-\phi)^n}{\sqrt{5}}$$

Let X be the set of all $n \in \mathbb{N}$ such that

$$a_n = \frac{\phi^n - (1-\phi)^n}{\sqrt{5}}$$

One can verify that a_1 and a_2 do indeed equal 1 using the formula above, thus $1 \in X$ and $2 \in X$. Now suppose that $k \in X$ for all $k < n$ where $n > 2$. By definition, we know that $a_{n-2} + a_{n-1} = a_n$, so

$$\begin{aligned}
a_n &= a_{n-1} + a_{n-2} \\
&= \frac{\phi^{n-1} - (1-\phi)^{n-1}}{\sqrt{5}} + \frac{\phi^{n-2} - (1-\phi)^{n-2}}{\sqrt{5}} \\
&= \frac{\phi^{n-1} - (1-\phi)^{n-1} + \phi^{n-2} - (1-\phi)^{n-2}}{\sqrt{5}} \\
&= \frac{\phi^{n-2}(\phi+1) - (1-\phi)^{n-2}(1-\phi+1)}{\sqrt{5}} \\
&= \frac{\phi^{n-2}(\phi+1) - (1-\phi)^{n-2}(2-\phi)}{\sqrt{5}} \\
&= \frac{\phi^{n-2}\phi^2 - (1-\phi)^{n-2}(1-\phi)^2}{\sqrt{5}} \\
&= \frac{\phi^n - (1-\phi)^n}{\sqrt{5}}.
\end{aligned}$$

Therefore, we conclude that $n \in X$ and that $X = \mathbb{N}$.

$\square$

Exercise 1.14

In this problem, you will prove some results about the **binomial coefficients**, using induction. Recall that
$$\binom{n}{k} = \frac{n!}{(n-k)!k!},$$
where n is a positive integer, and $0 \leq k \leq n$.

(a) Prove that
$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1},$$
$n \geq 2$ and $k < n$.

(b) Verify that $\binom{n}{0} = 1$ and $\binom{n}{n} = 1$. Use these facts, together with part (a), to prove by induction on n that $\binom{n}{k}$ is an integer, for all k with $0 \leq k \leq n$.

(c) Use part (a) and induction to prove the **Binomial Theorem**: For non-negative n and variables x, y,
$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k$$

Proof. (a) Using the recursive nature of factorials, that is
$$n! = n(n-1)!,$$
we deduce the following:
$$\begin{aligned}
\binom{n-1}{k} + \binom{n-1}{k-1} &= \frac{(n-1)!}{k!(n-k-1)!} + \frac{(n-1)!}{(k-1)!((n-1)-(k-1))!} \\
&= \frac{(n-1)!}{k!(n-k-1)!} + \frac{k(n-1)!}{k!(n-k-1)!(n-k)} \\
&= \frac{(n-k)(n-1)! + k(n-1)!}{k!(n-k-1)!(n-k)} \\
&= \frac{(n-1)!(n-k+k)}{k!(n-k)!} \\
&= \frac{n!}{k!(n-k)!} \\
&= \binom{n}{k}.
\end{aligned}$$

(b)
$$\binom{n}{0} = \frac{n!}{0!(n-0)!} = \frac{n!}{1 \cdot n!} = 1.$$
$$\binom{n}{n} = \frac{n!}{n!(n-n)!} = \frac{n!}{n! \cdot 1} = 1.$$

Let X be the set of all $n \in \mathbb{N}$ such that $\binom{n}{k}$ is an integer, where $0 \leq k \leq n$. One can manually verify that $1 \in X$ and $2 \in X$ (We need to verify that $2 \in X$ because the formula we'll use assumes $n \geq 2$). Now, assuming that $0 \leq k \leq n$, suppose that $\binom{r}{k}$ is an integer for all $r < n$ where $n > 2$. From (b), we know that $k = 0$ and $k = n$ both result in $\binom{n}{k}$ being an integer. We may now assume that $0 < k < n$. We know from our induction hypothesis that $\binom{n-1}{k}$ and $\binom{n-1}{k-1}$ are both integers, and we know from (a) that

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}.$$

Because $\mathbb{N}$ is closed under addition, $\binom{n}{k}$ must also be an integer. Therefore, $n \in X$ and $X = \mathbb{N}$.

(c) Let X be the set of all natural numbers n where the binomial theorem holds. We verify $n = 1$:

$$\begin{aligned}
\sum_{k=0}^{1} \binom{1}{k} x^{1-k} y^k &= \binom{1}{0} x^1 y^0 + \binom{1}{1} x^0 y^1 \\
&= 1 \cdot x + 1 \cdot y \qquad \text{Using part (a)} \\
&= x + y
\end{aligned}$$

Now suppose that $k \in X$ for all $k < n$ where $n > 1$. In particular, we deduce that the binomial theorem is true for $n - 1$. Thus,

$$\begin{aligned}
(x+y)^{n-1}(x+y) &= (x+y) \cdot \sum_{k=0}^{n-1} \binom{n-1}{k} x^{n-1-k} y^k \\
&= x \cdot \sum_{k=0}^{n-1} \binom{n-1}{k} x^{n-1-k} y^k + y \cdot \sum_{k=0}^{n-1} \binom{n-1}{k} x^{n-1-k} y^k \\
&= \sum_{k=0}^{n-1} \binom{n-1}{k} x^{n-k} y^k + \sum_{k=0}^{n-1} \binom{n-1}{k} x^{n-1-k} y^{k+1} \\
&= \sum_{k=0}^{n-1} \binom{n-1}{k} x^{n-k} y^k + \sum_{k=1}^{n} \binom{n-1}{k-1} x^{n-k} y^k \\
&=^{1} \left(x^n + \sum_{k=1}^{n-1} \binom{n-1}{k} x^{n-k} y^k \right) + \left(y^n + \sum_{k=1}^{n-1} \binom{n-1}{k-1} x^{n-k} y^k \right) \\
&= x^n + y^n + \sum_{k=1}^{n-1} \binom{n}{k} x^{n-k} y^k \qquad \text{Using part (a)} \\
&= \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k.
\end{aligned}$$

This closes the induction and shows that $X = \mathbb{N}$, as desired.

$\square$

[1] We evaluate the sums at $k = 0$ and $k = n$ to ensure that their indexes match and that they both sum to $n - 1$, this allows us to merge both of them into one summation.

Exercise 1.15

Criticize the inductive 'proof' showing that all cows are the same color.

Proof. The proof fails at $n = 2$. There's no way to guarantee the first cow picked has the same color as the rest of herd (which consists of 1 other cow). Indeed, if such guarantee existed, then the proof would work (Since you can guarantee ANY two cows in a herd have the same color, then the entire herd must be of the same color). $\square$

Exercise 1.16

Prove the converse of Theorem 1.1; that is, prove that the Principle of Mathematical Induction implies the Well-Ordering Principle. (This shows that these two principles are logically equivalent, and so from an axiomatic point of view it doesn't matter which we assume is an axiom for the natural numbers.)

Proof. Let X be an arbitrary non-empty subset of $\mathbb{N}$. Suppose, for the sake of contradiction, that X has no least element. Consider $Y = \mathbb{N} \backslash X$. We claim, and prove by induction, that $Y = \mathbb{N}$. That would inturn show that $X = \varnothing$ which would contradict our assumption that X is non-empty. Obviously $1 \in Y$, otherwise 1 would be the least element of X (there exists no $a \in \mathbb{N}$ such that $a < 1$). Now suppose that $r \in Y$ for all $r < n$ where $n > 1$. Suppose that $n \in Y$, but then n would be the least element of X (because $1 \leq r < n$ are all elements of Y, and thus they must not be elements of X). This contradiction forces us to conclude that $n \in Y$. We've met both demands required by the Principle of Mathematical Induction. Henceforth, $Y = \mathbb{N}$ and we conclude that X is empty, a contradiction. $\square$

Exercise 1.17

The *Strong* Principle of Mathematical Induction asserts the following. Suppose that X is a subset of $\mathbb{N}$ that satisfies that following two criteria:

(a) $1 \in X$, and

(b) If $n > 1$ and $n - 1 \in X$, then $n \in X$.

Then $X = \mathbb{N}$. Prove that the Principle of Mathematical Induction holds if and only if the Strong Principle of Mathematical Induction does.

induction $\implies$ *strong induction.* Let X be an arbitrary subset of $\mathbb{N}$ that satisfies both conditions of the Strong Principle of Mathematical Induction. In other words, $1 \in X$, and if $n > 1$ and $n - 1 \in X$, then $n \in X$. It suffices to show that $k \in X$ for all $k < n$ implies $n \in X$. Suppose that $k \in X$ for all $k < n$. In particular, that means $n - 1 \in X$, which implies $n \in X$ (Note that if $n = 1$, then the statement is vacuously true). Thus, using the Principle of Mathematical Induction, we infer that $X = \mathbb{N}$. This proves the Strong Principle of Mathematical Induction. $\square$

strong induction $\implies$ *induction.* Let X be an arbitrary subset of $\mathbb{N}$ that satisfies both conditions of the Principle of Mathematical Induction. In other words, $1 \in X$, and if $k \in X$ for all $k < n$, then $n \in X$. We already satisfy the first condition of the Strong Principle of Mathematical Induction. It suffices to show that $n - 1 \in X$ for $n > 1$ implies $n \in X$. To do that, we induct on the following set:

$$^{2}Y := \{a \in \mathbb{N} \ : \ r \in X \text{ for all } r < a\}.$$

Vacuously, $1 \in Y$. Now suppose that $k - 1 \in Y$ for $k > 1$. That means $r \in X$ for all $r < k - 1$, but we know from the properties of X that that implies $k \in X$. Hence, $r \in X$ for all $r < k$. So $k \in Y$. From the Strong Principle of Mathematical Induction, $Y = \mathbb{N}$. Going back to our induction on X, we know $n - 1 \in Y$. So $r \in X$ for all $r < n - 1$, and because $n - 1 \in X$, $r \in X$ for all $r < n$. From the properties of X, that implies $n \in X$. Henceforth, $X = \mathbb{N}$ follows from the Strong Principle of Mathematical Induction. $\square$

[2] Notice that we're performing two different inductions at the same time, the first is being performed on the set X, and the second is being performed on the set Y to prove a statement that we require to close our induction on X.